MSA680 - Lab 3

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1 Introduction

In this lab, we are performing survival analysis on data from 500 participants in the Worcester Heart Attack Study (Hosmer and Lemeshow, 2008) using R. Variables of interest from the original dataset are presented in Table 1 below:

Variable	Type	Description
LENFOL	Continuous	length of follow-up, terminated either by death or censoring
FSTAT	Categorical	censoring variable (loss to follow-up=0, death=1)
AGE	Continuous	age at hospitalization
BMI	Continuous	body mass index
HR	Continuous	initial heart rate
GENDER	Categorical	males=0, females=1

Table 1: Variables of Interest

For the next section, we will focus on only the 215 subjects who died during the study (i.e. FSTAT=1). The histogram in Figure 1 contains a first look at the data, where the length of follow up for only the subjects who died during the study is plotted. The histogram shows behavior that roughly matches an exponential distribution; this is expected, since heart-attack patients are most likely to die right after the event. The overlaid "smooth" density curve does not capture the behavior of the histogram near LENFOL=0 due to the fact that LENFOL is strictly non-negative, and the density curve does not have datapoints with negative LENFOL to interpolate accurately near 0. By dividing the cumulative number of patients who died by the total number of deaths, we can construct an empirical cumulative distribution function (cdf) of probability of death as shown in Figure 2 below. The probability of surviving less than 200 days is estimated to be $\frac{114}{215} \approx 53\%$, meaning that there is a large probability of death soon after event.

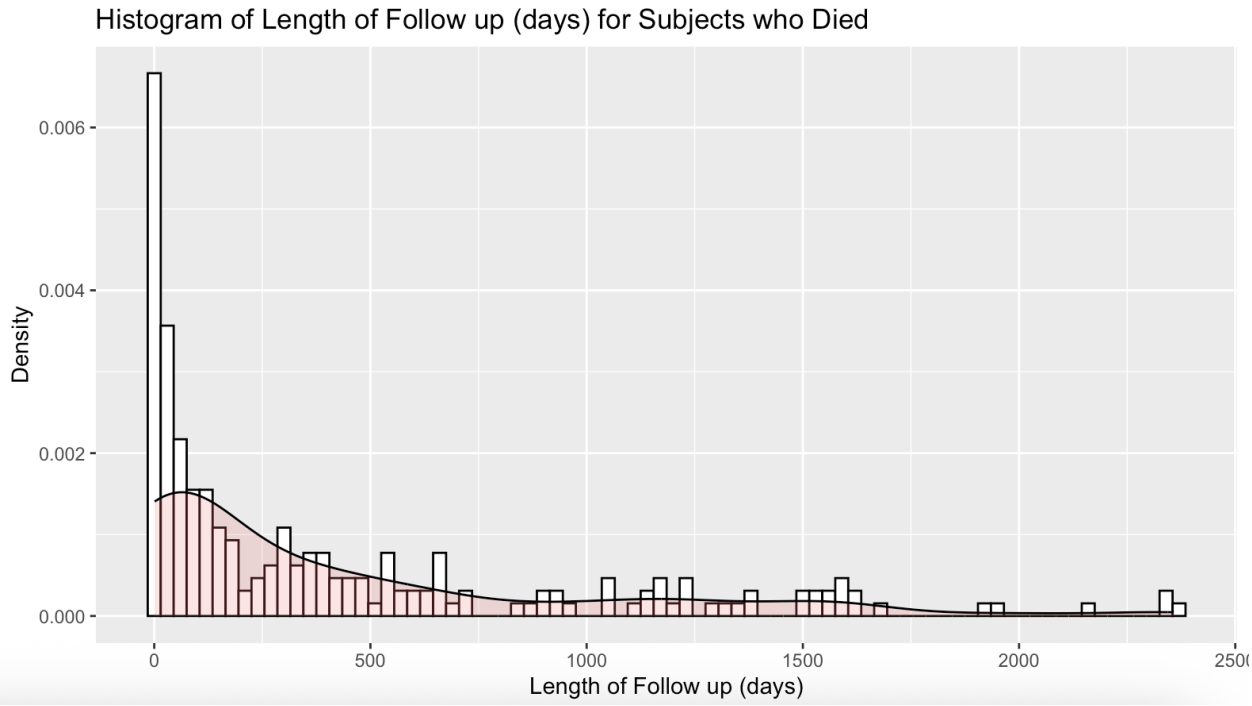


Figure 1: Histogram of Length of Follow-up (Subjects who Died during Study)

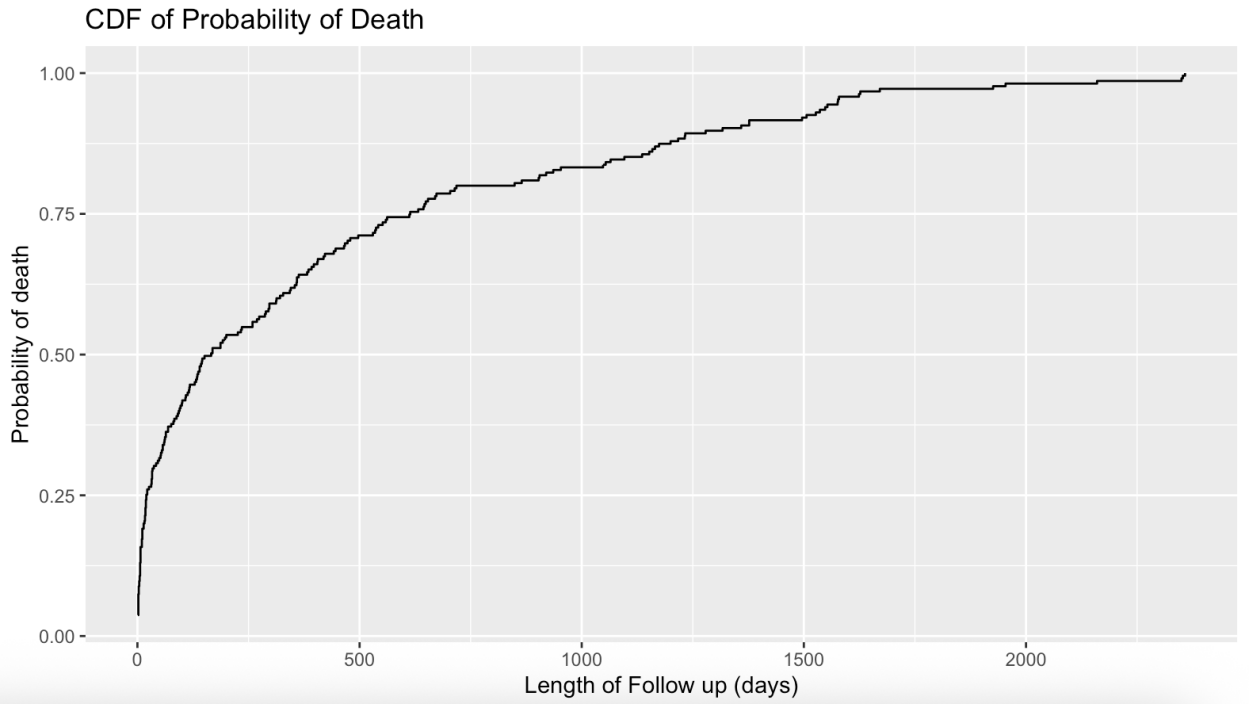


Figure 2: CDF of Probability of Death

A crude survival function can be estimated by computing $1 - CDF$, which is shown in Figure 3 below. As expected, there is a sharp decline right after the initial heart-attack event, and tapers off after the initial decline. The probability of surviving at least 1000 days is estimated to be $\frac{36}{215} \approx 16.7\%$, meaning that a fraction of patients who experience a heart-attack are estimated to survive for more than 1000 days.

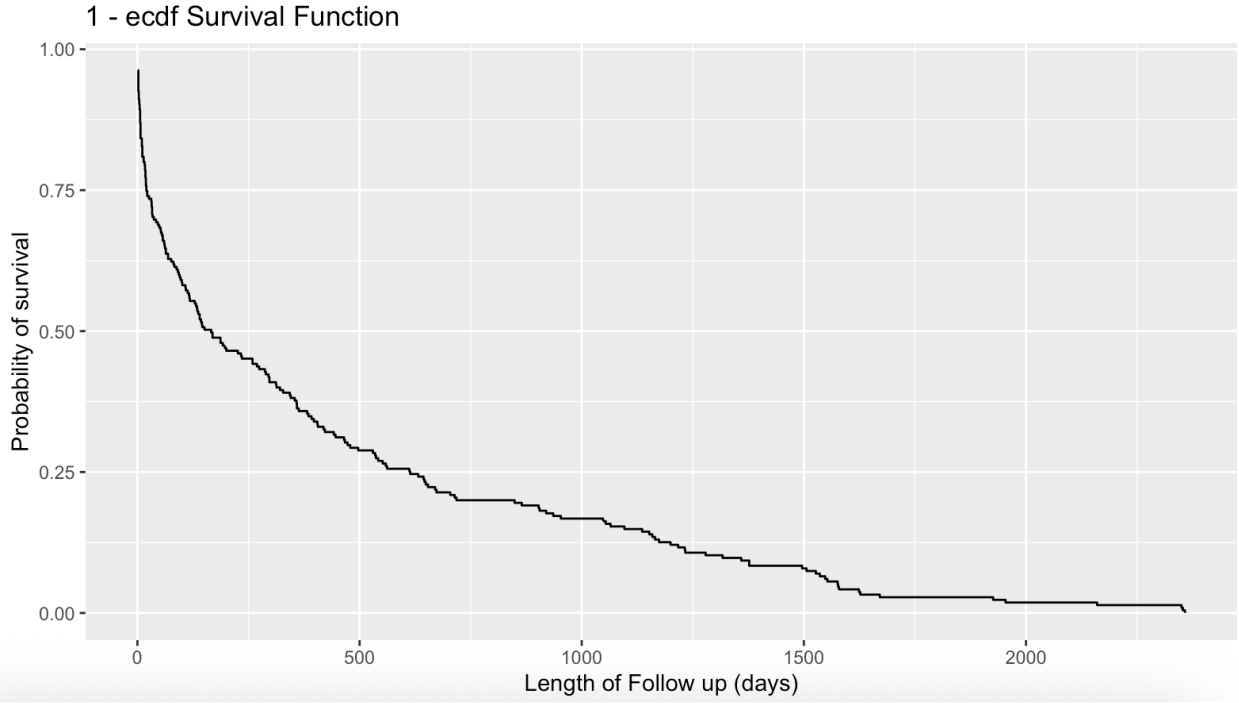


Figure 3: Probability of Survival

Now, we will look at summary statistics for the entire dataset – both patients who died and patients who were lost to follow up due to other reasons. The summary statistics are included in Table 2, and a "pairs" plot grouped by FSTAT is presented in Figure 4. A few observations are clearly noticeable: firstly, the average age at hospitalization was clearly higher for patients who died (77.9 years old, FSTAT=1) compared with those who were lost to follow-up (63.7 years old, FSTAT=0). Secondly, the length of follow up was also clearly higher for patients who died (avg. death at 425 days, FSTAT=1) compared with those who were lost to follow-up (avg. loss to follow up at 1228 days, FSTAT=0). Thirdly, for the patients who were lost to follow up (FSTAT=1), there were three main "waves" of times

when patients were lost to follow up (around 500 days, 1200 days, and 2000 days after heart attack) – perhaps these are determined by hospital protocol of when follow up stops, or perhaps due to the right-censoring of LENFOL data; this might be interesting to look into further. Not as obvious comparisons were heart rate and BMI; heart rate was slightly higher for patients who died, while BMI was slightly lower for patients who died, however it is not clear whether these differences are significant just from the graphs and table.

Variable	FSTAT	Subjects	Mean	StDev	Min	Max
LENFOL	0	285	1228	608	368	2192
	1	215	425	549	1	2358
AGE	0	285	63.7	13.4	30	90
	1	215	77.9	11.6	32	104
BMI	0	285	27.9	5.08	14.8	42.8
	1	215	24.9	5.35	13.0	44.8
HR	0	285	81.7	21.8	35	186
	1	215	94.1	24.1	36	157

Table 2: Summary Statistics Grouped by FSTAT (0=loss to follow up, 1=death)

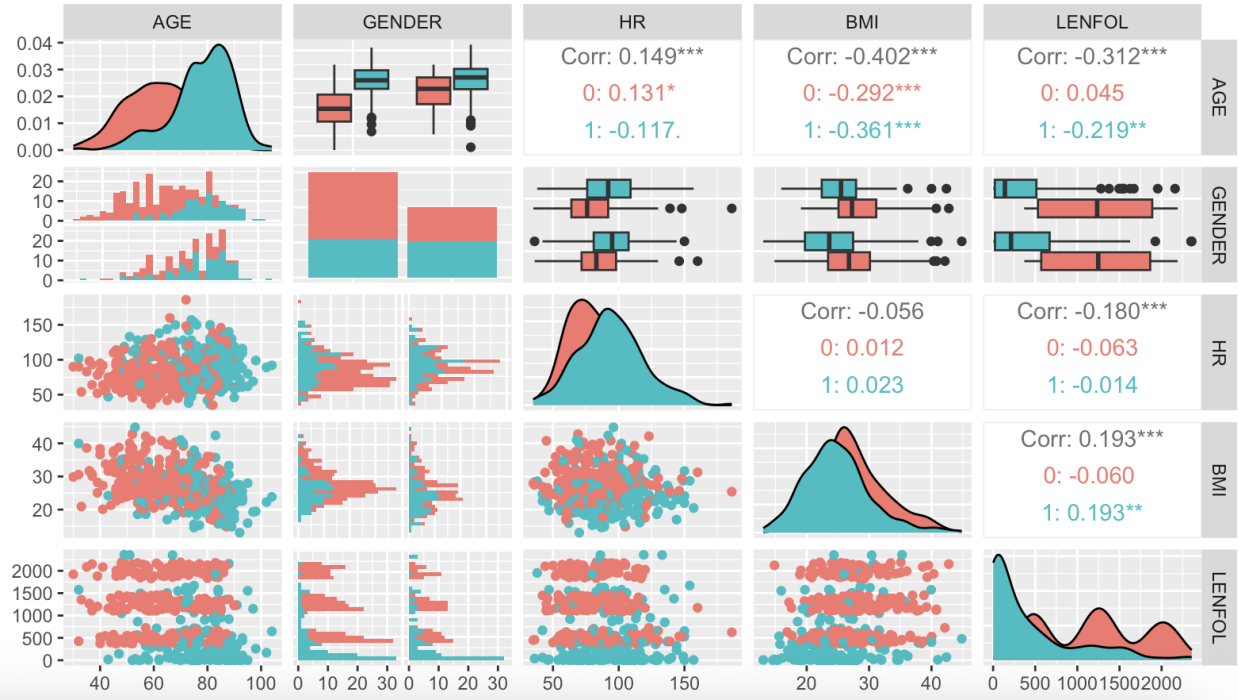


Figure 4: "Pairs" Plot Grouped by FSTAT (0=loss to follow up, 1=death)

2 Non-Parametric Analysis

Kaplan-Meier (KM) survival analysis allows for a more analytical approach to estimate the survival curve $S(t)$. It allows us to compare between categorical groups (e.g. gender), but not continuous or multivariate analysis due to its non-parametric nature. After fitting a KM survival function with no covariatates in the model (intercept only), select results are shown in Table 3 below.

Time	n_{risk}	n_{event}	Survival	Std. Err.	Lower 95% CI	Upper 95% CI
1	500	8	0.984	0.00561	0.9731	0.995
2	492	8	0.968	0.00787	0.9527	0.984
3	484	3	0.962	0.00855	0.9454	0.979
4	481	2	0.958	0.00897	0.9406	0.976
5	479	2	0.954	0.00937	0.9358	0.973
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1954	57	1	0.478	0.02970	0.4231	0.540
2160	12	1	0.438	0.04685	0.3552	0.540
2350	3	1	0.292	0.12325	0.1277	0.668
2353	2	1	0.146	0.12025	0.0291	0.733
2358	1	1	0.000	NaN	NA	NA

Table 3: Kaplan–Meier survival estimates (model with no covariates)

Looking at the KM estimates, we can see that now with censored patients in the model, the survival probabilities are very different from the crude 1-eCDF survival function from before. The median survival time is estimated to be $t = 1627$ days, and as we can see in the result summary, there is a sharp drop in survival probability after $t = 1954$ days due to the large decrease in the at-risk group due to the small number of participants remaining. The "plain" 95% confidence interval for the median survival time was $[1527, \text{NA}]$; the upper bound was calculated to be above 100% likely for the same reason – not enough data after $t = 1627$ for a sensible upper bound to be calculated. Setting `conf.type="log-log"` gives $[1506, 2353]$ as the 95% confidence interval for the median survival time; this condition restricts the CI bound limits to be at most 1, which allows for the upper confidence interval bound to be computed.

The survival function, as well as the cumulative hazard function for this model are graphed in Figures 5 and 6 below. Again, we can see that including censored patients in the KM survival estimate has made the KM survival and hazard curves very different from the crude eCDF survival estimates, as now the survival curve "flattens" out near 0.45 until the very end. We see again the three groups of censoring clearly present, which again is something to investigate further. The odd behavior after 2200 days is again due to the large decrease in the at-risk group due to the small number of participants remaining. The median survival time, estimated at $t = 1627$, indicates that for those who experienced a heart attack, 50% are expected to survive less than 1627 days after event, while 50% are expected to survive more.

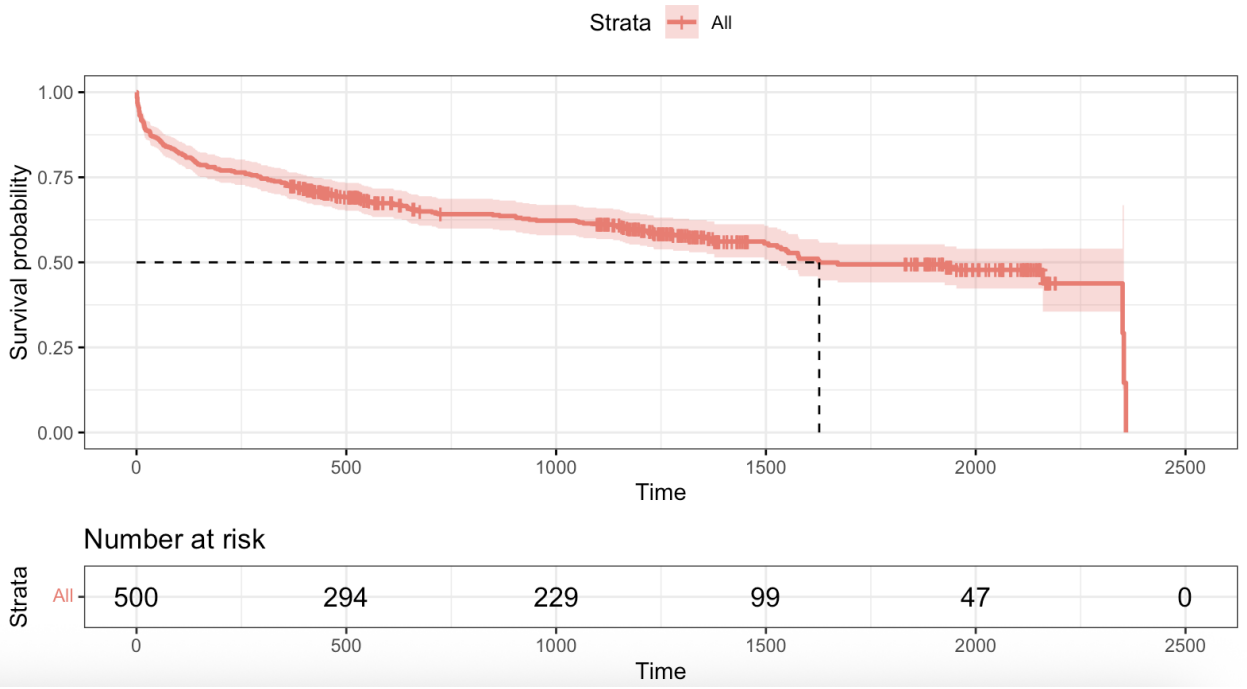


Figure 5: KM Survival Curve (intercept only model)

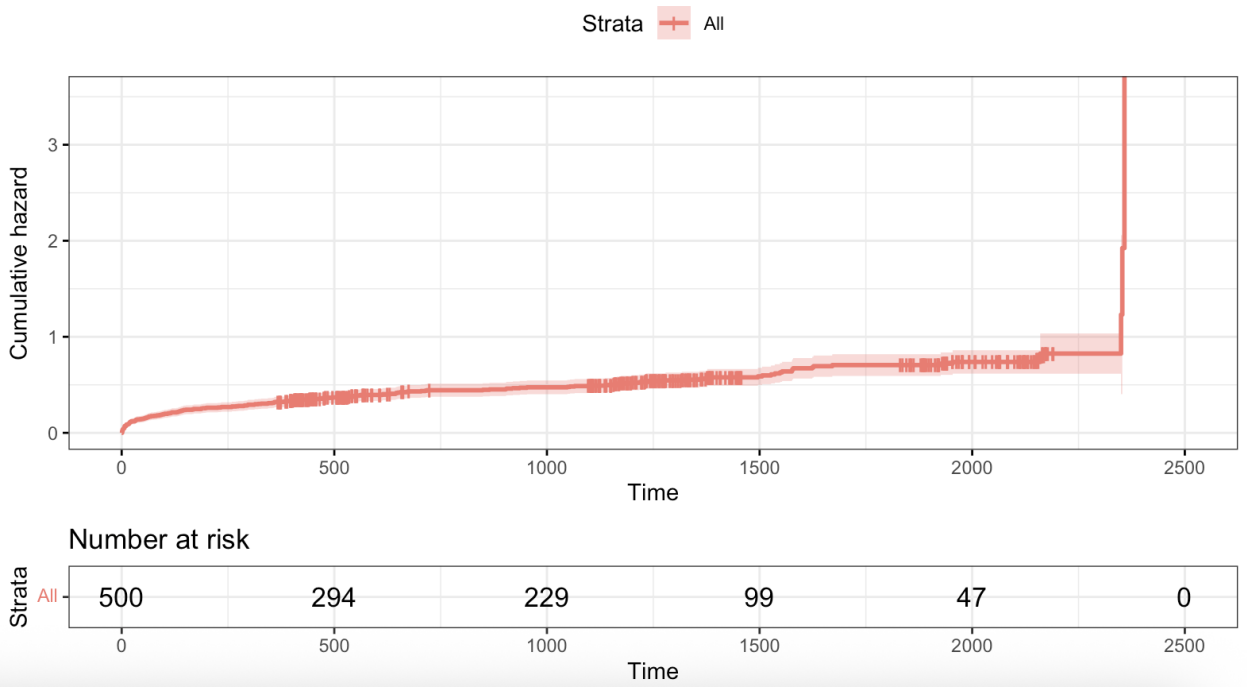


Figure 6: KM Cumulative Hazard Curve (intercept only model)

Since KM survival analysis allows for modelling separate survival functions for differing categorical covariates, we can perform the same analysis but with gender in the model. The resulting survival functions and hazard curves are shown in Figures 7 and 8 below. Visually, it seems that the survival curve is worse for females (green) compared with males (red), and the proportional hazards premise is fulfilled. Performing a logrank test to compare the survival between males and females gives a χ^2 test statistic of 7.8 with 1 degree of freedom, and the p value of $p = 0.005$ indicates that there is a significant difference between male and female survival at the 95% confidence level.

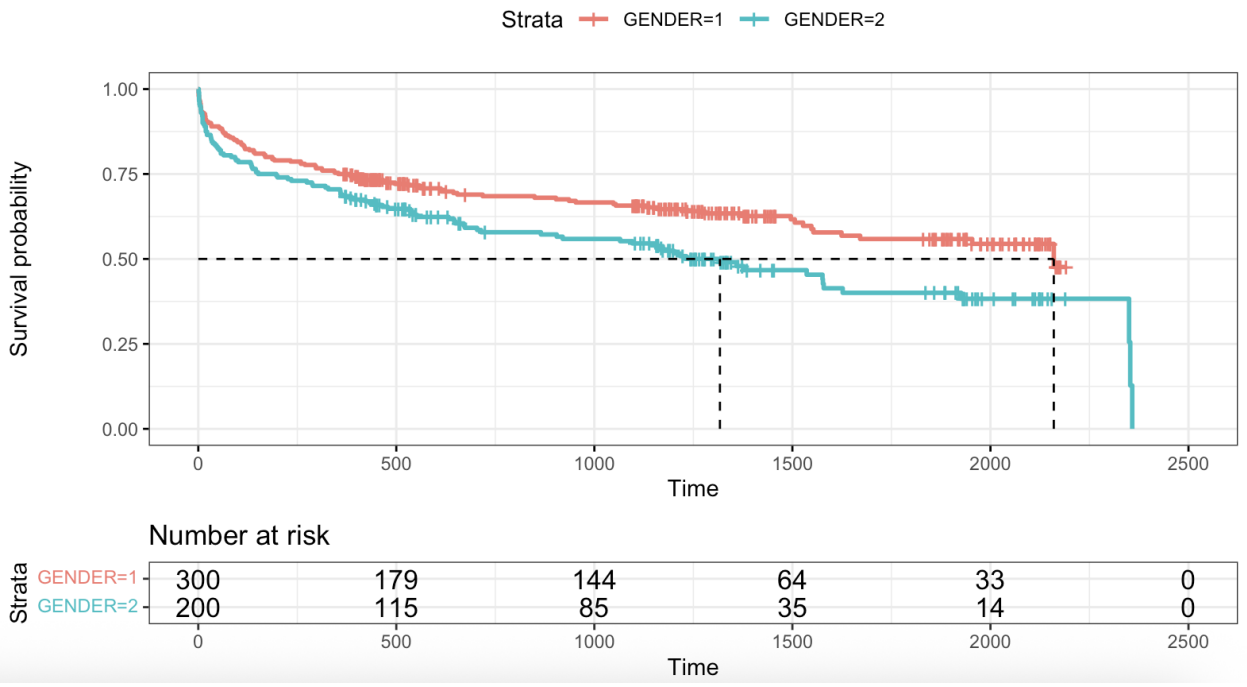


Figure 7: KM Survival Curve (model with gender covariate)

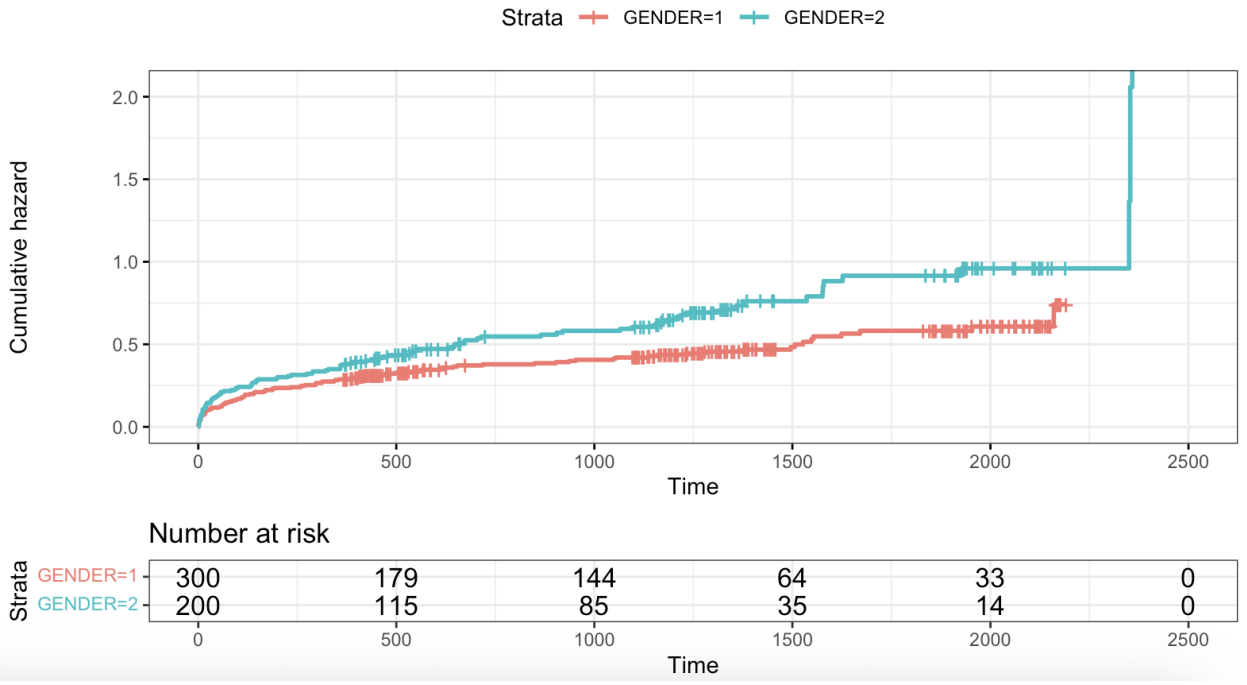


Figure 8: KM Cumulative Hazard Curve (model with gender covariate)

3 Parametric Analysis

Compared with KM survival analysis, the Cox Proportional Hazards (CoxPH) model allows for continuous multivariate analysis of survival due to its parametric nature. The underlying assumption is that the hazard functions are proportional for different groups. We can first consider a side example before returning to the WHAS dataset as illustrated by the dataset in Table 4:

Patient	**Survtime**	**Censor**	**Group**
1	6	1	C
2	7	0	C
3	10	1	T
4	15	1	C
5	19	0	T
6	25	1	T

Table 4: Side example: Patient survival data

Here, we have two groups – treatment and control. Assuming proportional hazard functions, the hazard function for subject i at failure time t_j can be defined as $h_i(t_j) = h_0(t_j)\psi_i$. $\psi_i = \exp(z_i\beta)$, where z_i is a 0/1 indicator for control and treatment respectively. For patients in the control group, ψ_i is expected to be $\exp(0) = 1$, while for patients in the treatment group, ψ_i is expected to be $\exp(\beta) = \psi < 1$ for some $\beta < 0$. The β parameter is to be estimated, and it is logical that we expect $\beta < 0$ since patients in the treatment group are expected to be less likely to experience the event at question. After deriving the partial log-likelihood function for ψ (Equation 1), we can then numerically optimize the optimal value for β as seen in Figure 9 below. Using R (see code in appendix), $\beta \approx -1.326129$ and $\log(L(\beta)) \approx -3.671981$.

$$
\begin{aligned}
L(\psi) &= p_1 p_2 p_3 p_4 \\
L(\psi) &= \frac{1}{3\psi+3} \cdot \frac{\psi}{3\psi+1} \cdot \frac{1}{1\psi+1} \cdot 1 \\
L(\psi) &= \frac{\psi}{(3\psi+3)(3\psi+1)(2\psi+1)} \qquad (1) \\
\log(L(\psi)) &= \log(\psi) - \log(3\psi+3) - \log(3\psi+1) - \log(2\psi+1) \\
\log(L(\beta)) &= \beta - \log(3e^{\beta}+3) - \log(3e^{\beta}+1) - \log(2e^{\beta}+1)
\end{aligned}
$$

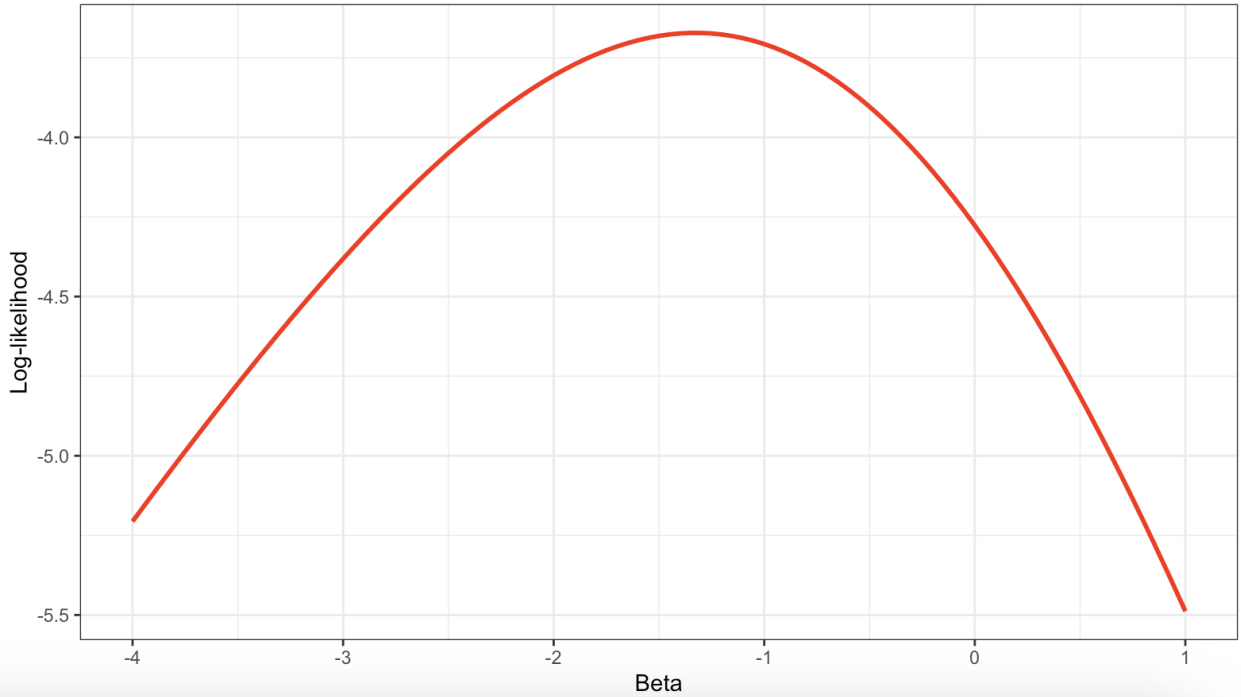


Figure 9: Maximizing $\log(L(\beta))$ for the side example

Using the `coxph` function directly in R, the output for the β estimate is -1.3261, which matches with the direct computations. This means that the hazard function for patients in the treatment branch is estimated to be $\psi = \exp(-1.3261) \approx 0.2655$ compared with the hazard function with patients in the control branch.

Going back to the WHAS dataset, we can apply the `coxph` function directly in R to fit a model with both gender and age (CoxPH allows for continuous variables, as well as

multivariate analysis). The results are summarized in Table 5 below. While the likelihood ratio test, Wald test, and Score (logrank) tests are all based on χ^2 test statistics, they test for different null hypotheses. The likelihood ratio test compares whether two nested models are significantly different from each other. In our case, the likelihood ratio test shows that the model with age and gender covariates is significantly different from a model with no covariates. The Wald test on the other hand determines whether the estimated effects are zero. For our case, the Wald test shows that the estimated effects of gender and age on the survival function are significantly different from zero. Finally, the logrank test is used to test the hypothesis of whether two survival functions (assuming proportional hazards) are significantly different from each other by using contingency tables at each event time. Again, the logrank test shows that the survival function for when gender and age are in the model is significantly different from the survival function with no parameters in the model. This is slightly surprising due to the differing conclusion from the KM survival analysis, which stated that gender did significantly affect survival when it was the only parameter in the model. However if we were to look at summary statistics for age grouped by gender, we will find that the 300 male participants had a mean age of 66.6 years, quite lower than the 200 female participants with a mean age of 74.7 years. Therefore after controlling for age, we can see that the effect of gender is no longer significant.

	Coef	Exp(Coef)	SE(Coef)	z	Pr($>\|z\|$)	95% CI
GENDER	-0.066285	0.935864	0.140585	-0.471	0.637	[0.7105, 1.233]
AGE	0.066928	1.069218	0.006196	10.802	$< 2 \times 10^{-16}$	[1.0563, 1.082]

Model Summary	Value	p
Concordance	0.731	
Likelihood ratio test (2 df)	142.4	$< 2 \times 10^{-16}$
Wald test (2 df)	119.7	$< 2 \times 10^{-16}$
Score (logrank) test (2 df)	126.9	$< 2 \times 10^{-16}$

Table 5: Cox proportional hazards model (age, gender covariates) results

At the 95% confidence level, we can see that the hazard functions are not significantly

different by gender, but are significantly different by age. That is, for participants at the same age, we cannot reject the null hypothesis that the male and female groups have the same hazard function. However, for participants of the same gender we were able to reject the null hypothesis that age has no effect on the hazard function, and for each 1 year increase in age, we expect the hazard function to be increased by a factor of 1.069. We can graph the survival functions by gender corrected for mean age, which is shown in Figure 10 below. Visually the curves are quite "close" to each other with overlapping confidence intervals; this matches with the parameter estimate results from before, which stated that gender effects were not significantly seen with age in the model.

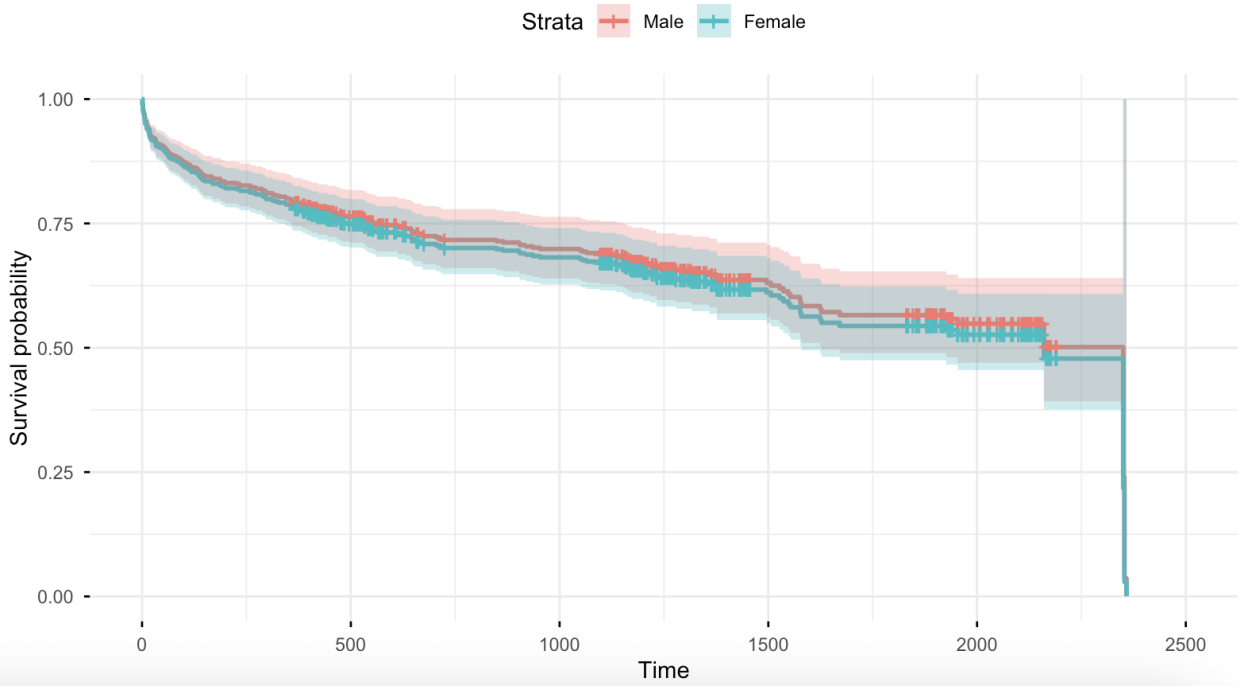


Figure 10: CoxPH Survival Functions by Gender (age adjusted at mean)

As the original dataset had many more covariates, we can try to fit a CoxPH model with more of the additional covariates to see if they have a significant effect on survival rates. The results for a model with all covariates (excluding DSTAT) are summarized in Table 6 below. At the 95% significance level, it seems that gender, age, heartrate, BMI, CHF (congestive heart failure), and SHO (shortness of breath?) significantly affected the hazard

function ratios. Fitting a CoxPH model with only the significant covariates stated, we can get results that are summarized in Table 7. We have now constructed three CoxPH models – the two covariate model (gender, age), the five covariate model (age, HR, BMI, CHF, SHO), and the full model (11 covariates). To compare the three models, the Akaike-Information Criterion (AIC) was calculated as 2316.276, 2260.716, and 2257.392 respectively. This shows that while the two-covariate model is significantly different from the other models, the five covariate model is not so different from the full model in terms of model fit, and would be the most parsimonious choice out of the three models examined.

Variable	Coef (β)	exp(β)	SE(β)	z	Pr($>\|z\|$)	Sig.
AGE	0.050252	1.051536	0.006707	7.493	6.73×10^{-14}	***
HR	0.008798	1.008836	0.002937	2.996	0.00274	**
SYSBP	-0.003970	0.996038	0.002092	-1.897	0.05779	.
BMI	-0.044905	0.956089	0.016347	-2.747	0.00602	**
CVD	-0.076106	0.926718	0.179346	-0.424	0.67131	
AFB	-0.017530	0.982622	0.176010	-0.100	0.92066	
SHO	1.105898	3.021937	0.282055	3.921	8.82×10^{-5}	***
CHF	0.749085	2.115065	0.149554	5.009	5.48×10^{-7}	***
AV3	0.290425	1.336995	0.430134	0.675	0.49955	
MIORD	0.113530	1.120226	0.149314	0.760	0.44705	
MITYPE	-0.345697	0.707727	0.190446	-1.815	0.06949	.

Table 6: Cox proportional hazards model (all covariates) results

Variable	Coef (β)	exp(β)	SE(β)	z	Pr($>\|z\|$)	Sig.
AGE	0.052637	1.054047	0.006576	8.005	1.20×10^{-15}	***
HR	0.009332	1.009375	0.002824	3.304	0.000953	***
BMI	-0.048718	0.952449	0.016044	-3.037	0.002393	**
SHO	1.011924	2.750889	0.261384	3.871	0.000108	***
CHF	0.723135	2.060883	0.146477	4.937	7.94×10^{-7}	***

Table 7: Cox proportional hazards model (age, HR, BMI, CHF, SHO covariates) results

4 Appendix – R Code

```
#Load libraries

library(tidyverse)

library(ggplot2)

library(GGally)

library(survival)

library(survminer)


# Reading in data

df <- read.csv(file="whas500.csv",header=T, sep=";")

df$FSTAT <- as.factor(df$FSTAT)

df$GENDER <- as.factor(df$GENDER)


# Plotting histogram (all subjects)

ggplot(df, aes(x=LENFOL)) +

  geom_histogram(aes(y=..density..), colour="black", fill="white", binwidth=30)+

  geom_density(alpha=.2, fill="#FF6666") +

  ggtitle("Histogram of Length of Follow up (days)") +

  labs(x="Length of Follow up (days)", y="Density")


# Plotting histogram (subjects who died)

ggplot(df[df$FSTAT == 1,], aes(x=LENFOL)) +

  geom_histogram(aes(y=..density..), colour="black", fill="white", binwidth=30)+

  geom_density(alpha=.2, fill="#FF6666") +

  ggtitle("Histogram of Length of Follow up (days) for Subjects who Died") +

  labs(x="Length of Follow up (days)", y="Density")
```

```
# Plotting cdf for death probability (dataset filtered for those who died)
ggplot(df[df$FSTAT == 1,], aes(x=LENFOL)) +
  stat_ecdf(geom = "step", pad=FALSE) +
  ggtitle("CDF of Probability of Death")+
  labs(x="Length of Follow up (days)", y="Probability of death")

# Calculating probability of surviving less than 200 days
df |> group_by(FSTAT) |>
  summarize(less_than_200 = sum(LENFOL<200), at_least_200 = sum(LENFOL >=200), n = n())
# So p=114/215, roughly 53%

# Plotting the 1 - ecdf survival function estimate
ggplot(df[df$FSTAT == 1,], aes(x=LENFOL)) +
  geom_step(aes(y = 1 - ..y..), stat = "ecdf", pad=FALSE) +
  ggtitle("1 - ecdf Survival Function")+
  labs(x="Length of Follow up (days)", y="Probability of survival")

# Calculating probability of surviving at least 1000 days
df |> group_by(FSTAT) |>
  summarize(less_than_1000 = sum(LENFOL<1000), at_least_1000 = sum(LENFOL >=1000), n = n
# So p=36/215, roughly 16.7%

# First look at the data
df %>% group_by(FSTAT) %>% summarise(N=n(),
                                     Mean=mean(LENFOL),
                                     SD=sd(LENFOL),
                                     Min=min(LENFOL),
```

```
                                        Max=max(LENFOL))

df %>% group_by(FSTAT) %>% summarise(N=n(),
                                     Mean=mean(AGE),
                                     SD=sd(AGE),
                                     Min=min(AGE),
                                     Max=max(AGE))

df %>% group_by(FSTAT) %>% summarise(N=n(),
                                     Mean=mean(BMI),
                                     SD=sd(BMI),
                                     Min=min(BMI),
                                     Max=max(BMI))

df %>% group_by(FSTAT) %>% summarise(N=n(),
                                     Mean=mean(HR),
                                     SD=sd(HR),
                                     Min=min(HR),
                                     Max=max(HR))

# Pairs plots
ggpairs(df, columns = c(2,3,4,7,18), ggplot2::aes(colour = FSTAT))

##########
# Non-parametric analysis
##########
df$FSTAT <- as.numeric(df$FSTAT)
```

```
df$GENDER <- as.numeric(df$GENDER)


# Creating survival object
surv.data <- with(df, Surv(LENFOL, FSTAT))


# Kaplan-Meier estimate
fit <- survfit(surv.data~1, data = df)
summary(fit)
# Median time at t=1627, SE=0.02822
# [0.4471, 0.558] for 95% CI probability of survival
# [1527, NA] for 95% CI probability of median time


# Kaplan-Meier estimate, log-log confidence interval
fit2 <- survfit (surv.data~1, data = df, conf.type="log-log")
fit2
# SE=0.02822, [0.4429, 0.553]


# Plotting the KM survival curve
ggsurvplot(fit,
           risk.table = TRUE,
           surv.median.line = "hv",
           ggtheme = theme_bw())


# Plotting the cumulative hazard curve
ggsurvplot(fit,
           fun = "cumhaz",
           risk.table = TRUE,
```

```
          ggtheme = theme_bw ())


# Separate KM curves by gender
fit3 <- survfit (surv.data~GENDER, data = df)
summary(fit3)
ggsurvplot(fit3,
           risk.table = TRUE,
           surv.median.line = "hv",
           ggtheme = theme_bw())


ggsurvplot(fit3,
           fun = "cumhaz",
           risk.table = TRUE,
           ggtheme = theme_bw ())


# Log-rank test to compare survival between genders
surv.test <- survdiff(surv.data~GENDER, data = df )
surv.test
# Significant difference at p=0.005


##########
#Parametric analysis
##########
# A random example
# Setting up the dataframe
random_df <- data.frame (
  Patient = seq(1, 6),
```

```
  Survtime = c(6, 7, 10, 15, 19, 25),
  Censor = c(1, 0, 1, 1, 0, 1),
  Group = c("C", "C", "T", "C", "T", "T")
)

# Constructing survival object, setting up Cox PH fit
random_surv.data <- with(random_df, Surv(Survtime , Censor))
random_fit <- coxph(random_surv.data~Group, data=random_df)
summary(random_fit)
# GroupT coef=-1.3261

#Computing by hand...
loglikelihood <- function(beta) {
  return(beta - log(3*exp(beta)+3)-log(3*exp(beta)+1)-log(2*exp(beta)+1))
}
ggplot(data.frame(x = c(-4, 1)), aes(x = x)) +
  geom_function(fun = loglikelihood, color = "red", size = 1) +
  labs(x = "Beta", y = "Log-likelihood") +
  theme_bw()

result <- optim(par = 0, fn = loglikelihood, method = "L-BFGS-B",
                control = list(fnscale = -1),
                lower = -4, upper =1)
# Beta partial likelihood estimate = -1.326129, same as above!
# Value of partial log-likelihood = -3.671981
```

```
# Going back to the WHAS dataset...
fit4 <- coxph(surv.data~GENDER + AGE , data = df)
summary(fit4)
# Gender not significant, but age is

# Correcting by mean age, then graphing separate regression curves
sex.dt <- with(df,
               data.frame(GENDER = c("Male", " Female " ),
                          AGE = rep(mean(AGE, na.rm = TRUE), 2)))

surv.fit.gender <- survfit (fit4, data = df, newdata = sex.dt)

ggsurvplot(surv.fit.gender,
           conf.int = TRUE,
           legend.labs = c("Male","Female"),
           ggtheme = theme_minimal())

# Trying all covariates in a full model...
# DSTAT excluded since wasn't sure whether it was death stat...
fit5 <- coxph(surv.data~ AGE + HR + SYSBP + BMI + CVD + AFB + SHO +
                 CHF + AV3 + MIORD + MITYPE, data = df)
summary(fit5)
# Seems like age, hr, BMI, SHO, CHF significant...

fit6 <- coxph(surv.data~ AGE + HR + BMI +
                SHO + CHF, data = df)
summary(fit6)
```

```
extractAIC(fit4)

extractAIC(fit5)

extractAIC(fit6)
```